Predicting NBA Player Salaries: A Multi-Method Statistical Analysis

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NBA salaries are truly puzzling: How can one player earn $30 million while another player putting up similar numbers is getting a fraction of that? What are these top paid players doing that others aren't? I performed advanced modeling methods, including Correspondence Analysis, Factor Analysis, and several regularized regression models to try and answer that exact question.

I used a dataset of 7,298 NBA players which included general demographics such as team, position, age, along with numerical statistics such as baskets made, rebounds, minutes played and more. Position immediately gave me trouble as out of the 7,298 players, 113 of them were listed as playing 2 distinct roles such as PF-SG. I decided to remove these positions and just kept the first position, which was the player's primary position. The dataset included data spanning from 2010 to 2025 and was almost exclusively numerical which meant I had to create categorical variables based on slices of the numerical total. After creating these, I was able to take a closer look as to how different categories of players relate to one another and salary levels by employing Correspondence Analysis.

Correspondence analysis can be thought of as a map where things that belong together will appear closer to each other, and vice versa. As well as this, in drawing a line from a certain variable that I wanted to compare through the origin, I can look at the distance of each point relative to that line, with points closer to the line having profiles that are proportional to or strongly aligned with the reference point's profile: they share the same pattern of deviation from the average. Fundamentally, CA is about ratios and proportions and helped visually guide thoughts and ideas as I proceeded into further analysis. Looking at **Figure 1**, I was able to see that position matters for pay. Centers and Power Forwards cluster near the higher salary tiers, while Shooting Guards and Point guards are more spread across the lower salary levels. This does not mean these players are less valuable, just that the market historically has paid a premium for strong interior players. Secondly, I was able to find that age does not easily predict salary. Veteran, Prime, and Young players all showed no strong pull toward any particular tier. More experience does not necessarily mean more pay in the NBA. Third, I found that assists are closely tied to salary, which led me to consider that playmaking ability might be something that NBA franchises actively look for in a player.

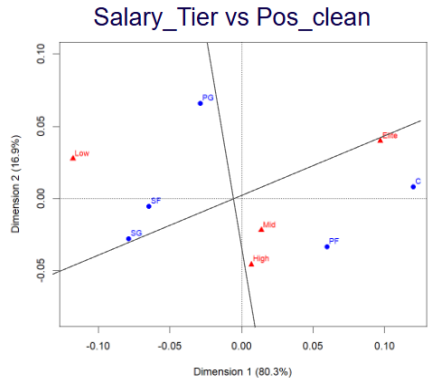

Figure 1

Using correspondence analysis, I was able to get a much clearer starting point for the deeper statistical work that will follow. By understanding which player characteristics are associated with salary level, I was able to build more informed models to actually predict what a player should earn. By understanding which player characteristics are associated with each other, I was able to proceed into Factor Analysis with more valuable factors to see if I can crack the NBA salary code.

By using Factor Analysis, which is an analysis technique to decrease the number of variables that are in a model to make it more interpretable, I was able to group over twenty individual box score statistics into four border performance dimensions instead of analyzing each stat separately. The results showed that there are clear categories in place with those being: scoring volume (minutes, shot attempts, and total points), rebounding (offensive and defensive boards), perimeter shooting volume (three-point attempts and makes), and scoring efficiency (shooting percentages like FG% and eFG%). These four factors explain most of the variation in a player's performance and also confirm that NBA statistics are usually clustered into meaningful groups. However, when I used these grouped performance factors to predict salary, they explained only a very small portion of salary differences, suggesting that pay in the NBA depends on more than just measurable box score production.

To understand what contributes the most to an NBA player's paycheck, I used a statistical model known as Elastic Net. In the context of the data, Elastic Net takes the specified predictors I want it to look at and determines which stats actually matter to a player's salary and which ones are unnecessary and could be disregarded. Elastic Net provided the following output, seen in **Figure 2**. The numbers on the

right-hand side show how one additional unit of a stat contributes to a salary. For example, age has a coefficient of 457,483, and this means that for every one additional year a player is, the model predicts a player's paycheck will increase by around $457K. You might notice that some of the stats have a dot next to them, such as steals or field goal attempts, which indicates that the model considered these to be unhelpful for the salary prediction. The model may have discarded these because they could be redundant since other shooting variables such as 3-pointers attempted already count the times a player tries a shot or the model may have decided they barely add any value to a player's contract. Some of the lucrative stats for players included whether the player was a starter, adding about $1.7M to their average earnings, playing as a center or power forward, adding around $1M, as well as blocks, contributing another $1M. Age also adds around another $500K. On the other hand, personal fouls is considered to be a huge negative costing a player over $1M and turnovers costing a player an estimated $220K. Overall, this output shows what the NBA market rewards such as starting status, experience seen with age, and rim protection with blocks and views stats such as fouls and turnovers in a negative light when considering a player's worth.

It's important to note that in every model that I made to predict salary, there were "core four" pillars that were always significant in predicting salary. These are if a player is a starter, points, blocks, and assists. While each model might yield something different, the core four pillars allow me to come to the conclusion that a holistic view of the players is necessary. The core four pillars encompass offensive statistics, defensive statistics, playmaking, and any player's time on the court.

	lambda.1se
(Intercept)	-11809715.94
Pos_cleanSG	.
Pos_cleanSF	352949.17
Pos_cleanPF	922228.88
Pos_cleanC	982740.46
Age	457483.51
G	-19120.92
GS	.
MP	-164920.30
FGA	.
FG.	.
X3P	377590.91
X3PA	534750.24
X3P.	-1565345.29
X2PA	.
X2P.	1513486.30
eFG.	.
FTA	129507.85
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ORB	22339.81
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BLK	1188730.23
TOV	-219703.65
PF	-1033725.11
PTS	606366.23
Is_Starter	1673681.82

Figure 2

Introduction

This project focuses on analyzing NBA player salaries using statistical modeling techniques., Basketball was selected due to strong domain interest and data availability. It's always fascinating to see the insane amounts that the NBA organizations are able to pay their players, especially the first round picks or franchise legends, and this was the exact thought I had in deciding to create a model that would use player statistics to predict how their salaries are decided. While an immediate thought could be the player who scores the most points will get paid the most, more often than not, there is more that decides who is worth it. In this report, I will be performing 4 lines of analysis in an attempt to create a model that can most accurately explain the variance in salary of an NBA player.

Exploratory Analysis:

The dataset that I have done lines of analysis for and investigated, involved an NBA dataset with players as the observations. The dataset included stats from 7,298 players from 2010-2025. The initial dataset included over 37 variables, including demographics, shooting, defense, efficiency, and contract details. The target variable for all of the analyses conducted was the Salary variable that indicates how much a player is paid annually. The original dataset had many missing values in the various shooting variables due to the player not attempting this shot type. These missing values were addressed by inputting a 0 in place of them to be able to run my analyses. A majority of the predictors that are seen in the dataset are numeric; however, I found a few predictors that could be transformed into meaningful categorical variables to be able to run correspondence analysis. Some of the predictors that were created include Scoring_Tier, which categorizes players into scoring buckets ranging from a bench warmer to classifying them as a key contributor. Next, Age_Group was another predictor that was created to categorize players by age, including young, prime, and veteran. Lastly, the Is_Starter binary predictor was created that classified NBA players as a 1 if they were on the starting lineup for >=50% of their games and a 0 otherwise. Upon looking at Pos (player position), there was a clear issue that needed to be solved: Some players (131 total observations out of 7,298) have 2 positions, such as SG-PF, which created a lot of noise that needed to be worked out. This led to me deciding to only include each player's primary role for all further analysis, which is denoted by the new variable, **Pos_clean**.

Residual Analysis/Model Diagnostics:

Continuing on with the exploratory analysis, I looked at the target variable of salaries and noticed that this predictor is heavily skewed to the right. This represents the reality of the NBA, where there are a few well-known superstar players who are being paid a supermax contract, while a majority of players are being paid a benchwarmer salary. While looking at the residual plots, I noticed that there was a heavy funnel shape in the residuals, indicating heteroscedasticity. The normality plot also does not appear linear, so I decided to conduct a transformation of the data entirely with a log transformation. After the transformation, the funnel shape that appeared in the original dataset drastically improved, although there was still a slight funnel shape. The normality plot also saw some improvements, particularly in the supermax earners of the NBA. This shows the trend between high-performance rookies who are on a fixed first-year contract, compared to veterans whose contracts are often based on their playing stats. I also took a look at the VIF diagnostics to understand where multicollinearity occurred. This also helped me understand

Figure 1

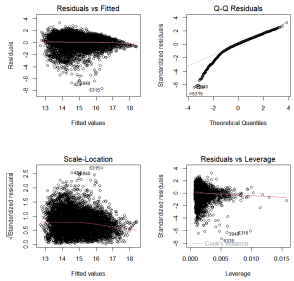

Figure 2

	GVIF	Df	GVIF^(1/(2*Df))
Pos_clean	4.359787	4	1.202079
Age	1.098357	1	1.048025
G	2.428685	1	1.558424
GS	7.723978	1	2.779204
MP	12.837506	1	3.582947
eFG.	1.341550	1	1.158253
FTA	5.931408	1	2.435448
FT.	1.399821	1	1.183140
ORB	3.686032	1	1.919904
DRB	5.339755	1	2.310791
AST	5.900279	1	2.429049
STL	2.779649	1	1.667228
BLK	2.381477	1	1.543204
TOV	7.196938	1	2.682711

which predictors to include when running the analyses. From the VIF's that were outputted, I understood that the field goals attempted and percentages from 2-point, 3-point, and free throws had heavy amounts of multicollinearity present. I also found that minutes played and Is_Starter were heavily correlated with one another. With all of this in mind, I determined that this model provides sufficient data to be able to apply various lines, and I was also able to determine the most important predictors when trying to learn more about the data and formulate practical conclusions.

Correspondence Analysis:

Prior to model construction, I employed Correspondence Analysis to further assist in exploratory techniques by helping to characterize associations between categorical player attributes that something like a heatmap or corrplot cannot do. By reducing the contingency table of two attributes into a low-dimensional visual space, where the proximity of row and column points reflects the strength and direction of their association, I can get a better idea of the variables that work together to predict Salary across NBA players. All of the Correspondence Analyses are performed using the ca package in R with visualizations produced through base plot methods.

After creating Scoring_Tier and Age_group, I realized there was more to be done with the variables, thus several new predictors were created: Salary_Tier, which split the Salary variable into 4 tiers (Low, Mid, High, Elite), MP_3, which divided Minutes played into 3 buckets (Low, Medium, High), and AST_3, which divided Assists into the same 3 quantiles (Low, Medium, High). Using these newly created features, I can now apply the analyses to see how they relate to one another, and most importantly how they correlate to salary. The results of the correspondence analysis plots are as shown:

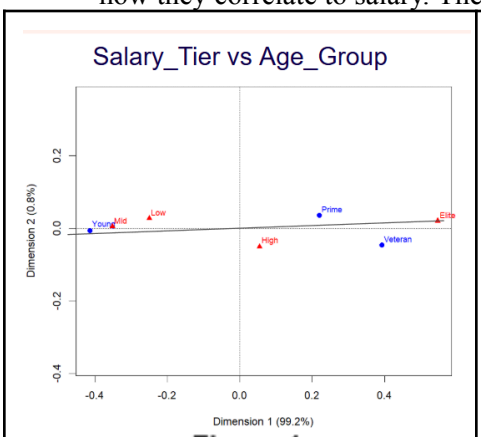

Figure 1

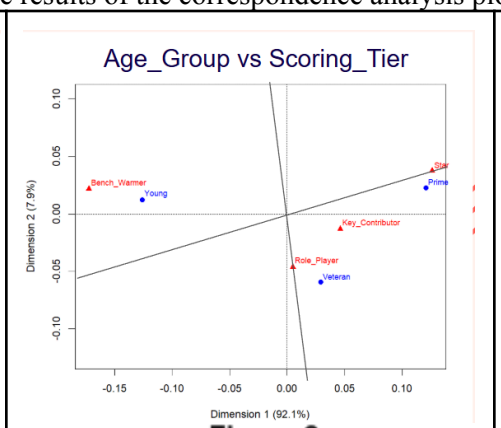

Figure 2

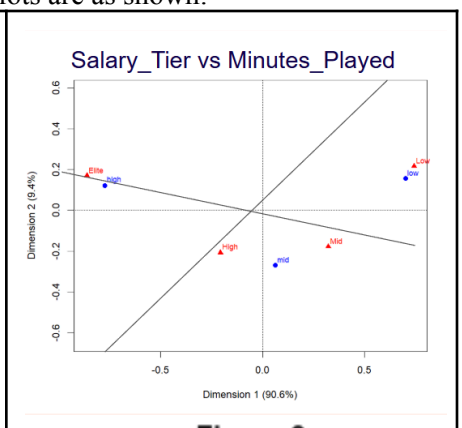

Figure 3

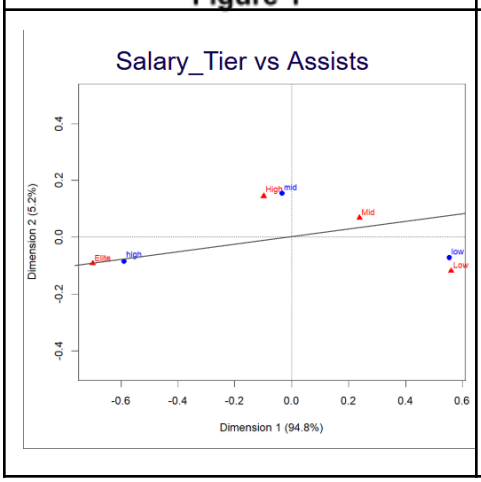

Figure 4

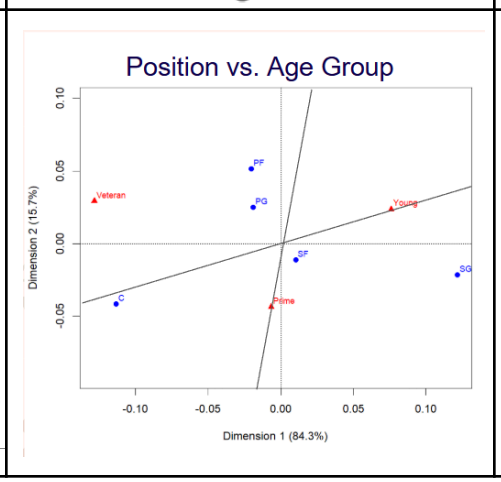

Figure 5

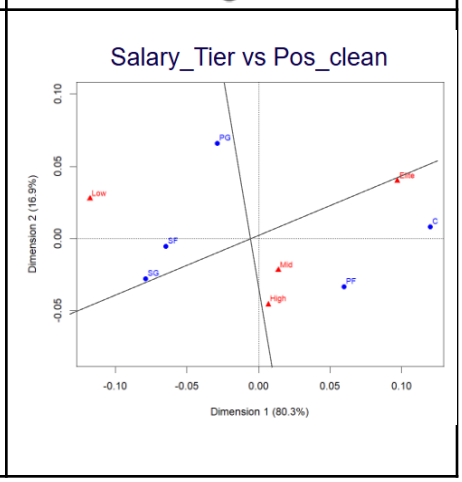

Figure 6

A total of 12 distinct Correspondence Analyses were conducted covering combinations of Position, Salary, Age, Scoring Tier, Minutes Played and Assists. Some of the most important analyses that gave me something to build off as I proceeded into model selection are denoted in the table above.

Key Findings

The CA of Position vs. Salary (Figure 6) captures 97.2% of total inertia across the two dimensions, indicating it is almost a complete representation of the association structure between these two variables. Dimension 1 (80.3%) corresponded to a salary gradient increasing from left to right with C (Center) loading strongly positive and SG (Shooting Guard) loading strongly negative. SF (Small Forward) is closest to the origin, meaning it is the closest to an "average salaried" role that I have. Dimension 2 revealed a secondary separation between Point Guards and Shooting Guards that Dimension 1 alone could not capture as both of these positions leaned towards a lower salary while their salary profiles differed in structure.

The CA of Age vs. Salary (Figure 1) showed no strong relational pull between age cohorts and their salary tier. The absence of a clear gradient can suggest that experience and career stage alone are not good enough predictors of earnings in the NBA, and that some players might just have that 'special something'. This will be a good test of the models as I proceed forward to see if they can capture what that special something just might be.

The CA of Scoring Tier vs Salary (not pictured) gave me the expected association: Star players mapping tightly to Elite salaries and Benchwarmers to Low salaries. This helps serve as an internal validity check to confirm that the salary tiering was meaningfully constructed.

The CA of Assists vs. Salary and Minutes Played vs. Salary showed that both of these features had positive relationships with salary. This confirmed that playmaking metrics and amount of time spent on the court carry salary signal and reinforced their relevance as predictors for further stages of analysis.

Further Usage

The natural next step for these newly created variables is to turn them into ordinal factors and apply them in factor analysis to see how the quantiles might be more valuable or more easily interpretable as factors rather than plain numerical variables. The most notable relationship to potentially discover would be that performance dimensions could very well be position-dependent which would greatly help my analysis as different positions have different requirements and statistics that might not be easily translated into regular regression models.

Regularized Regression:

All Subsets

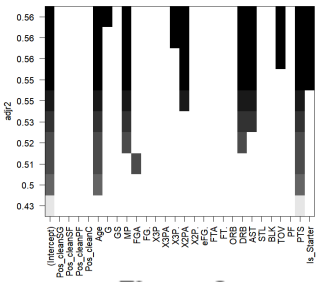

Figure 3

I started the analyses by running various forms of regularized regression, including All Subsets, Ridge, LASSO/Relaxed LASSO, and Elastic Net. Starting off with All Subsets regression, I restricted the model to only look at 10 predictors since this technique is computationally expensive. If I were to look at all of the predictors in the original dataset as well as the predictors I created, the R program would have likely crashed in the process. After running this, the results can be seen in **Figure 3.** The top performing models achieved an R-squared of around .56, with the high impact performance predictors being points, assists, defensive rebounds, and blocks. The age, Is_Starter, 3-point percentage, 2-pointers attempted, and minutes played all turned out to be significant predictors as well. One important observation I noted is that the Is_Starter predictor appeared in almost every high-performing subset, indicating that it is a strong predictor when predicting NBA players' salaries.

Ridge

When looking at basketball data, it's often said that a holistic view of the player is important to determine their impact, thus their salary. Other regularized regression techniques remove variables that are insignificant, but I believe that it's important to give some weight to every variable, which is exactly what ridge regression does. First it was important to look at the k-fold cross-validation plot to determine what lambdas to use. At the minimum lambda value, the ridge model yielded an R^2 value of 0.5937. This is higher than previously seen in the Ordinary-Least Squares Regression R^2 value of about 0.56. Running ridge regression at the one standard error lambda value gave a R^2 of 0.5804. It's important to note that the R^2 value indicates how much of the data is captured with the model that's calculated, but it still does not give an accurate idea of the predictive power that the ridge model has. In this case, it especially doesn't give a clear idea of what is a strong model since the R^2 values are very close.

To get more information on what is a strong prediction model, the RMSE was calculated at the different lambda values. At the minimum lambda value, the RMSE calculated is 5598473. This is about on par with the rest of the models. At the 1se lambda value, the RMSE was 5689547, which is a bit higher than at the lambda min. Based on the R^2 and RMSE values, the ridge regression model using the minimum calculated lambda value is a strong predictor. The model is shown below:

Salary = −754,834,100 + 368,521.8 × Year + 513,638.6 × Age − 11,691.54 × G + 3,938.642 × GS − 101,574.8 × MP + 496,728.1 × FG + 20,713.44 × FGA + 5,460,784 × FG% + 1,122,925 × X3P + 283,123.2 × X3PA − 2,179,995 × X3P% + 477,086.6 × X2P − 83,606.21 × X2PA − 1,455,050 × X2P% − 5,511,859 × eFG% + 449,906.3 × FT + 276,014.6 × FTA − 673,277.6 × FT% − 364,185.8 × ORB + 449,869.0 × DRB + 212,152.5 × TRB + 736,756.2 × AST − 159,299.1 × STL + 1,123,279 × BLK − 185,077.3 × TOV − 990,095.9 × PF + 166,701.4 × PTS + 1,701,784 × Is_Starter + 164,700.9 × Volume_Shooter − 62,797.22 × isWest + 62,496.24 × isEast − 443,716.1 × isSG − 815,727.2 × isPG + 497,836.1 × isC + 210,543.3 × isSF + 725,658.1 × isPF

Looking at the values, the eFG%, X3P%, and X2P% have large negative coefficients. However, FG%, Is_Starter, Blocks, and X3P have very large positive coefficients. This can be indicative that a percentage of shots made is not as important as raw points contributed throughout a season for an NBA player. This is also due to percentage being on a scale from 0-100, with most players being less than 50%. This is still considered really good, but players that don't shoot so much but still have an impact on the court are covered by the positive indicators of Is_Starter, Blocks, etc.

LASSO

Moving on from Ridge, I also conducted another regularized regression technique that is able to predict a target variable, in this case, being salaries, by selecting the most important predictors and shrinking the less important variable coefficients to exactly 0. This method is effectively performing automatic variable selection, which is one positive of this method compared to Ridge. The model output from LASSO can be seen in **Figure 4** and achieved an R-squared value of around .558, meaning that the model is explaining around 55.8% of the salary variation. The RMSE achieved from this model was around $5.68M, which represents the average prediction error of the salary. Some of the biggest positive impact predictors observed were the Is_Starter binary predictor, contributing around $1.6M to a player's salary. This was closely followed by the 2-point percentage predictor at $1.3M, blocks contributing $1.2M, defensive rebounds at $930K, assists at $790K, and points with $670K. The largest negative contributors to a player's salary were personal fouls, decreasing the predicted salary by

	lambda.1se
(Intercept)	-11819495.089
Pos_cleanSG	.
Pos_cleanSF	274233.588
Pos_cleanPF	809444.703
Pos_cleanC	862568.804
Age	461505.161
G	-18161.653
GS	.
MP	-178328.410
FGA	.
FG.	.
X3P	.
X3PA	610102.803
X3P.	-1439453.343
X2PA	-2894.329
X2P.	1269311.022
eFG.	.
FTA	.
FT.	-527986.563
ORB	.
DRB	926851.620
AST	790759.973
STL	.
BLK	1151994.031
TOV	-170186.589
PF	-973400.008
PTS	670460.777
Is_Starter	1640912.218

Figure 4

around $970K, and turnovers at around $170K. Both of these observations seem to make reasonable sense since a higher value in both turnovers and personal fouls is associated negatively with a player's performance and may impact how a team views that player, further impacting how much they are willing to pay that player. Furthermore, LASSO excluded several predictors to reduce the noise and overfitting of the model by shrinking the shooting guard position, field goals attempted, field goals made, 3-point percentage, efficiency, offensive rebounds, steals, etc. Another important observation this output highlighted is that the power forward and center positions contribute the most to salary relative to the point guard position. Traditionally, these players are also some of the tallest on the floor, so this may potentially hint at the idea that taller players are likely to be paid more; however, deeper analysis is needed to be done to confirm this observation.

Relaxed LASSO:

After refitting the model that was taken from LASSO, I refit it to create a sparser model using relaxed LASSO. The model that was selected kept 17 variables, and had an R^2 value of 0.5595. This indicates a strong predictive power in comparison to the other methods of regularized regression that were used for this dataset. The Residual Standard Error is $5,566,000, which is also lower than seen in the other models that I had looked at. However, as seen in the other models Is_Starter is the strongest positive predictor on salary, adding roughly $2.28 million. Points per game and blocks also are strong positive predictors, even larger than assists. It is important to note that Age is a strong positive predictor, which can be seen in the correspondence analysis graphs as well. Compared to LASSO, it's important to note that the relaxed LASSO model did not keep the 3 point shooting variables, which strengthens the understanding that 3-point shooting is not strongly correlated with a higher salary. All of the models reinforce the idea that salary is increased by a holistic view of a player, which is increasingly evident in the models keeping position variables as well as both offensive and defensive variables.

Elastic Net

Lastly, the final type of regularized regression that I conducted was Elastic Net. This type of regression is useful because it can combine a mix of Ridge and LASSO properties. The results that I received from this model were very similar to the previous regularized techniques I had used and can be seen in **Figure 5**. The R-squared was .558 with an RMSE of $5.6M. The biggest positive contributors were once again the Is_Starter predictor at $1.7M, blocks at $1.2M, and defensive rebounds at $915K. In terms of scoring impact, some of the highest contributing predictors were 2-point percentage, contributing $1.5M, assists at $826K, and points at $600K. Looking at the positional contributions with point guards as the baseline, small forwards contribute around $350K, power forwards contribute $920K, and centers contribute $980K. These findings also suggest the idea that taller positions tend to have higher earnings relative to point guards since these three positions are typically the tallest players for an NBA team. The negative contributors were the same with personal fouls and turnovers, along with the dropped features of steals, field goals attempted, efficiency, and the shooting guard position.

	lambda.1se
(Intercept)	-11809715.94
Pos_cleanSG	.
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Pos_cleanPF	922228.88
Pos_cleanC	982740.46
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TOV	-219703.65
PF	-1033725.11
PTS	606366.23
Is_Starter	1673681.82

Figure 5

After running all of these variations of regularized regression, I had come to the conclusion that the Elastic Net model was the optimal model choice. Unlike LASSO and Ridge, Elastic Net retained a majority of the volume metrics in the data. It also provided a more realistic view of the spacing on the court by including more positional impacts. Elastic Net was also able to address the multicollinearity between stats such as points, assists, and defensive rebounds without losing individual skill set nuances. As a result of Elastic Net, I came to the conclusion of the "core four" pillars to investigate further, as a majority of these predictors were consistent throughout the models and provide a good summary of a player and what their earnings will likely be. This "core four" consists of the Is_Starter predictor, points,

blocks, and assists. These will be investigated further as I head into more analyses later in this paper, such as correspondence and factor analysis, which will work on dimension reduction of this dataset.

Quantile Regression:

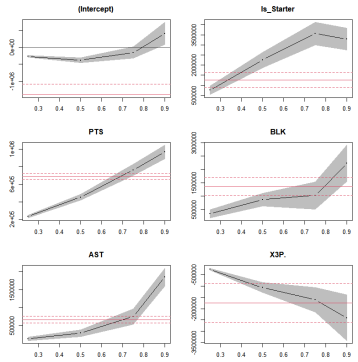


Figure 6

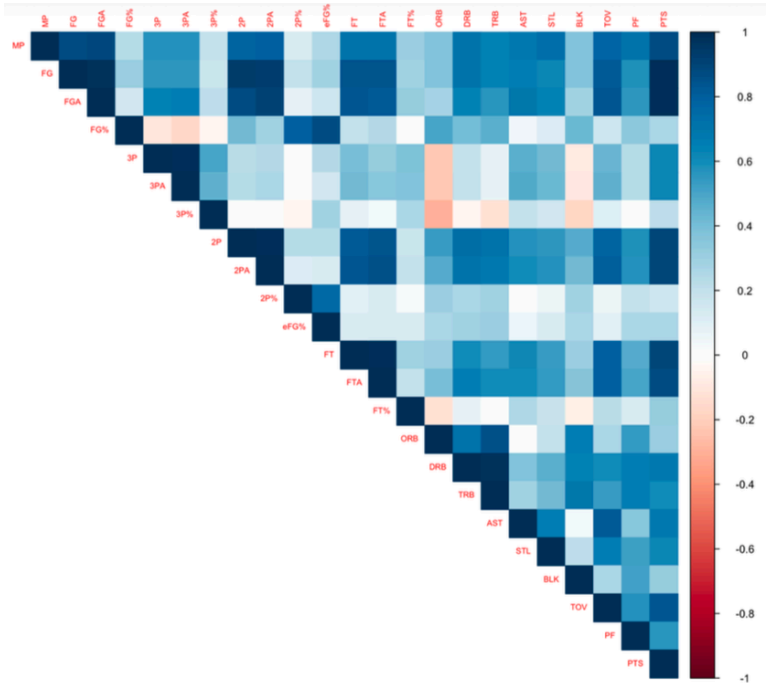
While regularized regression can output the contribution of certain predictors to the salary for an average NBA player, the "average" player does not truly exist in the NBA market. Since the NBA salary market is heavily skewed right, the average values are not resistant to outliers, and the average contributions a stat has to a player are being pushed up. To combat this issue, I decided to look at conditional quantiles that represent various salary tiers of NBA players and see how "valuable" a stat is to a player of that salary quantile. I looked at the 25th percentile, which represents a role player's salary, the 50th percentile, representing a median salary, the 75th percentile, highlighting the higher-end salary, and the 90th percentile, showcasing superstar players on a supermax contract. Quantile Regression has many benefits, including being able to handle heteroscedasticity, as seen in the residual analysis I conducted earlier. It is also robust to outliers and minimizes the impacts of overpaid and underpaid players. It is also an exceptionally good route to go if the data is a bit nonlinear, which was seen with the normality plot. Some of the major findings I found with Quantile Regression are that the Is_Starter variable is worth much more for supermax contract players compared to role players. This predictor contributes $3M to supermax players while adding $587K to role player salaries. Assists also have an exponential growth with value, as it contributes $120K for role players and $1.3M to supermax players. Moreover, points seem to have a linear pattern as the salary increases, and shooting efficiency goes down as the salary increases. This may indicate that a team prioritizes a player's ability to create shots and be a playmaker compared to being good at catch-and-shooting skills, as they want players who can run the floor and set the tempo for a team. Overall, Quantile Regression is a great way to address some of the gaps seen in regularization and can show how stats impact players of varying salaries, given the nature of the high skewedness of NBA salaries.

Factor Analysis:

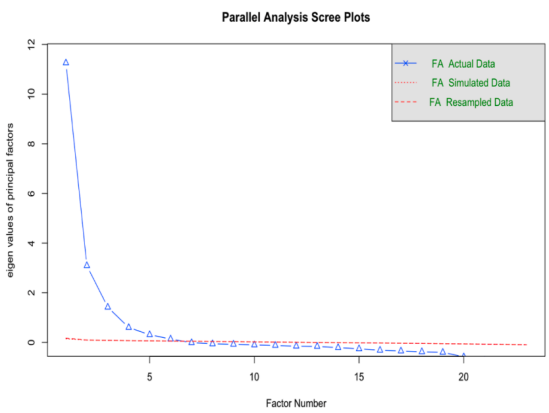
Quantile regression helped show how certain statistics impact salary at different points of the pay scale. I wanted to take a step back and better understand how the performance statistics themselves are structured. Many of the box score variables move together, and when you look at them individually in regression models, it can create redundancy and overlap. Instead of analyzing each stat separately, it makes more sense to identify broader performance dimensions that summarize these relationships. This is where factor analysis comes into play. This allows me to reduce the large number of correlated statistics into a smaller set of meaningful underlying factors before examining how they are related to salary.

To begin, I created a correlation heatmap of all 23 performance variables to visually examine how the statistics relate to one another. The goal was to confirm if there were strong correlation patterns before running factor analysis. In the heatmap, you clearly see large blocks of dark blue along scoring-related variables. Minutes played (MP), field goals (FG), field goal attempts (FGA), two-point attempts (2PA), free throws (FT), and total points (PTS) are all strongly positively correlated. This confirms that these variables move together and likely measure the same underlying concept: offensive volume. Rebounding stats like offensive rebounds (ORB), defensive rebounds (DRB), and total rebounds (TRB), clearly group together as well. Efficiency matrics like FG%, 2P%, and eFG% do not align with volume stats. This suggests that they measure something different. This confirmed that natural groupings exist in the data.

Next, I used a scree plot with parallel analysis to determine how many factors to keep. The scree plot shows a sharp drop after the first factor and begins to flatten after the fourth. The parallel analysis indicates that only the first four eigenvalues are above the simulated line. Because of this, retaining four factors made statistical sense.

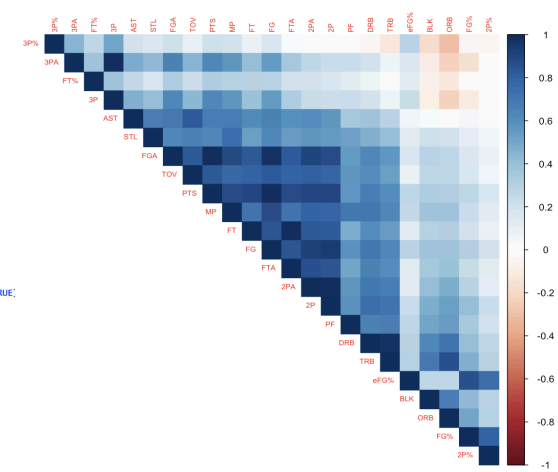


After varimax rotation, the structure became clear. Factor 1 loads heavily on FG, FGA, 2P, 2PA, FT, MP, and PTS, representing scoring volume. Factor 2 loads on ORB, DRB, and TRB, representing rebonding. Factor 3 loads on 3P and 3PA, capturing perimeter shooting volume. Factor 4 loads on FG%, 2P%, and eFG%, representing scoring efficiency. Together, the four factors explain about 77% of the total variance. Scoring volume, rebounding, and efficiency are all statistically significant but 3-point volume isn't. The R^2 is only 0.008 which means the factors explain less than 1% of the salary variation.



Even though the four-factor solution was supported by the scree plot and parallel analysis, I wanted to briefly examine what would happen if you extracted five factors instead. Since the fifth eigenvalue was close to the cutoff point, it was reasonable to check whether adding one more factor would meaningfully change the structure or improve the variance explained.

When I ran the five-factor model with varimax rotation, the overall structure remained very similar. The first factor still loads heavily on FG (0.873), FGA (0.871), 2P (0.897), and TRB (0.897) loading strongly



BEFORE (4 factor)

```
Standardized loadings (pattern matrix) based upon correlation matrix
      ML1   ML3   ML2   ML4   h2     u2 com
MP   0.72  0.39  0.37  0.07 0.82 0.1825 2.1
FG   0.89  0.29  0.30  0.16 1.00 0.0033 1.5
FGA  0.89  0.23  0.39  0.01 0.99 0.0073 1.5
FG%  0.16  0.25 -0.14  0.92 0.96 0.0368 1.3
3P   0.30  0.00  0.95 -0.02 0.99 0.0056 1.2
3PA  0.33  0.00  0.93 -0.09 0.98 0.0196 1.3
3P%  0.06 -0.15  0.51  0.12 0.30 0.6963 1.3
2P   0.92  0.34 -0.05  0.19 1.00 0.0042 1.4
2PA  0.95  0.30 -0.04  0.07 0.99 0.0112 1.2
2P%  0.03  0.16  0.01  0.80 0.67 0.3330 1.1
eFG% 0.01  0.17  0.26  0.93 0.96 0.0442 1.2
FT   0.82  0.22  0.16  0.04 0.76 0.2447 1.2
FTA  0.82  0.29  0.09  0.06 0.76 0.2365 1.3
FT%  0.24 -0.08  0.33  0.03 0.17 0.8272 2.0
ORB  0.23  0.78 -0.31  0.24 0.82 0.1812 1.7
DRB  0.48  0.85  0.06  0.14 0.97 0.0308 1.6
TRB  0.43  0.88 -0.06  0.18 1.00 0.0045 1.5
AST  0.64  0.04  0.29 -0.04 0.50 0.4999 1.4
STL  0.54  0.22  0.26  0.00 0.41 0.5863 1.8
BLK  0.23  0.62 -0.16  0.23 0.52 0.4821 1.7
TOV  0.79  0.24  0.21 -0.01 0.72 0.2815 1.3
PF   0.43  0.52  0.11  0.15 0.49 0.5057 2.2
PTS  0.87  0.26  0.38  0.12 0.99 0.0120 1.6

                       ML1  ML3  ML2  ML4
SS loadings           8.26 3.73 3.12 2.65
Proportion Var        0.36 0.16 0.14 0.12
Cumulative Var        0.36 0.52 0.66 0.77
```

AFTER (5 factor)

```
> print(fa_model$loadings, cutoff = 0.4, sort = TRUE)

Loadings:
     ML1    ML3    ML2    ML5    ML4
MP    0.694  0.423
FG    0.873
FGA   0.871
2P    0.897
2PA   0.926
FT    0.876                       0.406
FTA   0.865
AST   0.645
STL   0.533
TOV   0.788
PTS   0.870
ORB          0.796
DRB   0.449  0.863
TRB          0.897
BLK          0.633
PF    0.405  0.542
3P                0.952
3PA               0.930
3P%               0.512
FG%                      0.922
2P%                      0.801
eFG%                     0.926
FT%

               ML1   ML3   ML2   ML5   ML4
SS loadings    8.150 3.965 3.136 2.641 0.381
Proportion Var 0.354 0.172 0.136 0.115 0.017
Cumulative Var 0.354 0.527 0.663 0.778 0.794
```

```
Loadings:
     ML1    ML3    ML2    ML5    ML4
MP    0.694  0.423
FG    0.873
FGA   0.871
2P    0.897
2PA   0.926
FT    0.876                       0.406
FTA   0.865
AST   0.645
STL   0.533
TOV   0.788
PTS   0.870
ORB          0.796
DRB   0.449  0.863
TRB          0.897
BLK          0.633
PF    0.405  0.542
3P                 0.952
3PA                0.930
3P%                0.512
FG%                       0.922
2P%                       0.801
eFG%                      0.926
FT%

                 ML1   ML3   ML2   ML5   ML4
SS loadings    8.150 3.965 3.136 2.641 0.381
Proportion Var 0.354 0.172 0.136 0.115 0.017
Cumulative Var 0.354 0.527 0.663 0.778 0.794
```

together. The perimeter shooting factor is still clearly defined by 3P (0.952) and 3PA (0.930). Efficiency also remains distinct, with FG% (0.922), 2P% (0.801), and eFG% (0.926) loading strongly onto the same factor.

The fifth factor explains a much smaller portion of variance (around 1.7%), which increases the cumulative variance from about 77% to roughly 79%. Though this explains a slight increase in total variance, it does not fundamentally change the interpretation of the performance dimensions. The main four categories remain dominant, and the additional factor does not introduce a dramatically new performance construct.

I also recognized the correlation heatmap using the AOE ordering method. This does not change the correlation values themselves, but it reorders the variables to cluster highly correlated variables more cleanly. In the updated heatmap, the scoring volume variables, rebounding cluster, perimeter shooting, and efficiency measures appear more tightly grouped along the diagonal. This makes the underlying structure of the data more visually apparent and further supports the factor groupings identified in both the four and five factor solutions. Checking the five-factor model confirms that the core performance structure is stable and reinforces that the original four-factor solution was already capturing the primary dimensions.

Conclusions and Practical Significance

After running and analyzing all lines of analyses, I was able to gain an understanding of the most important predictors when predicting an NBA salary. Almost all of the analyses indicated that a player's starting status, position, experience, and overall playmaking abilities were considered to be some of the most important indicators for salary. The analyses I conducted helped successfully reduce the number of dimensions in this highly complex dataset and formulate an interpretable understanding behind the NBA salary market. However, depending only on the box score statistics can create a skewed perspective on the reality of the NBA. The "average" player is inherently difficult to define in any sports league as there are so many players that fall outside of the range that is considered "normal" for salaries. To achieve a highly successful model, outside factors that might not appear in typical box scores such as a player's reputation, injury history, ability to work with other teammates, and specific team needs play a huge part in salary negotiations. In terms of significance, these conclusions could play an important part in facilitating conversations in NBA teams front offices in deciding what a fair player compensation should be. It could also be used for deciding whether a player's contribution and performance is actually worth the amount they are being paid and can be used as a sort of performance review.

Takeaways

The biggest takeaways from this project was understanding a dataset may not tell a complete story on its own and transformations or other techniques might be needed to understand and create a stronger model with more predictive power. It is important to acknowledge the importance of having someone with subject matter to fully understand the results and understand the "why" behind results. Additionally, I noted the importance of not just basing predictions on averages, but understanding and trying to find patterns between those that fall in and out of the norm to gain a greater grasp of a dataset. Finally, I learned the importance of parsimony in modeling as more does not always mean better. More data can create confusing conclusions to explain to others, while a simple, effective dataset can be the most impactful when explaining to others and trying to incorporate the results of a model into the real world.